

1. Val. funkcija, verjetnost, Schrödingerjeva enačba

- Poznamo pred SE: • delec opišemo kot valovanje, $\lambda = \frac{2\pi}{k}$ in $v = \frac{\omega}{k}$
• $E = \hbar\omega$, $p = \hbar k$, $E = \frac{p^2}{2m}$

- valovna enačba: $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$, $u = u_0 \cos(kx - \omega t)$

$$\rightarrow -k^2 = -\frac{\omega^2}{c^2} \Rightarrow \hbar\omega = \pm \hbar ck \Rightarrow E \propto p$$

- Difuzijska enačba: $D \frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}$, $T = T_0 e^{i(kx - \omega t)}$

$$\rightarrow D = \frac{i\hbar}{k^2} = i \frac{\hbar}{2m} \Rightarrow E = \frac{p^2}{2m}$$

- umetno dodamo še člen $V(\vec{r})$ in enačbo posplošimo v 3D

$$\rightarrow \text{Schrödingerjeva enačba} \quad i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi$$

Val. funkcija $\Psi = \alpha + i\beta$, $\alpha, \beta, V \in \mathbb{R}$, vstavimo v SE

$$\rightarrow -\hbar \dot{\beta} = -\frac{\hbar^2}{2m} \alpha'' + V\alpha \quad \text{in} \quad -\hbar \dot{\alpha} = -\frac{\hbar^2}{2m} \beta' + V\beta$$

Bornova interpretacija: Ψ^2 je verjetnostna gostota za detekcijo delca v volumnu dV , Ψ je verjetnostna amplituda.

$\vec{r}$ ni koordinata, ampak ker lahko vemo le verjetnost, da pri $\vec{r}$ detektiramo delec.

$$\text{Velja} \quad \int |\Psi|^2 d^3\vec{r} = 1, \quad \rho = |\Psi|^2$$

$$\text{Kont. en.} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0, \quad \text{velja} \quad \frac{d}{dt} |\Psi|^2 = \frac{\partial \Psi^*}{\partial t} \Psi + \Psi^* \frac{\partial \Psi}{\partial t}$$

$$\rightarrow \text{odvoda izrazimo iz SE} \Rightarrow \frac{d}{dt} |\Psi|^2 = - \underbrace{\frac{\partial}{\partial x} \left(\frac{\hbar}{2im} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \right)}_{\vec{j}} - \underbrace{\frac{2}{\hbar} \text{Im}(V) |\Psi|^2}_0$$

$$\text{Pričakovana vrednost} \quad \langle x \rangle = \int \Psi^* x \Psi d^3\vec{r}$$

$$\text{Hitrost težišča porazdelitve:} \quad v_T = \frac{d\langle x \rangle}{dt} = \dots = \frac{1}{m} \int \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi d^3\vec{r}$$

→ en integral gre proti 0

2. Diracov zapis, Hilbertov prostor, operatorji

$$\hat{x} = x, \quad \hat{p} = -i\hbar \frac{\partial}{\partial x}, \quad \hat{x}^n = x^n, \quad \hat{p}^n = (-i\hbar)^n \frac{\partial^n}{\partial x^n}, \dots$$

$$\hat{f}(z) = \sum_n c_n z^n \rightarrow \text{funkcija operatorja} \quad \hat{f}(\hat{A}) = \sum_n c_n \hat{A}^n. \quad \text{Če velja } \hat{A}\psi = a\psi: \hat{f}(\hat{A})\psi = f(a)\psi \quad \rightarrow \text{lasten vektor}$$

• v kv.mech. vsaki merljivi vrednosti in opazljivki pripada operator

$$\text{Hamiltonov operator} \quad \hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}, \quad \hat{V} = V(x, t)$$

$$\text{Komutator: } [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

$$\cdot [\lambda \hat{A}, \hat{B}] = \lambda [\hat{A}, \hat{B}]$$

$$\cdot [\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$$

$$\cdot [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \quad (\text{vrstni red!})$$

$$\cdot \text{Jacobi: } [\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$$

$$\cdot \text{Baker-Hausdorff: } e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \frac{1}{3!} [\hat{A}, [\hat{A}, [\hat{A}, \hat{B}]]] + \dots$$

$$[x, p] f(x) = -i\hbar \frac{\partial f}{\partial x} + i\hbar f + i\hbar \frac{\partial f}{\partial x} = i\hbar f, \quad \text{3D: } [x_\alpha, p_\beta] = i\hbar \delta_{\alpha\beta}$$

$$\frac{d\langle \hat{A} \rangle}{dt} = \dots = \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle$$

$$\text{Simetričen (hermitski) operator: } \langle \psi | \hat{A} | \psi \rangle = \langle \psi | \hat{A} \psi \rangle = \langle \hat{A} \psi | \psi \rangle, \quad \text{npr. } \frac{\partial^2}{\partial x^2}$$

$$\text{Anti-sim. (Anti-herm.) operator: } \langle \psi | \hat{A} \psi \rangle = -\langle \hat{A} \psi | \psi \rangle, \quad \text{npr. } \frac{\partial}{\partial x} \quad \text{--- uporabimo per partes}$$

$$\text{Hermitsko adjungiran operator: } \langle \psi | \hat{A} \psi \rangle = \langle \hat{A}^\dagger \psi | \psi \rangle$$

$$\text{Sebiadjungiran operator: } \hat{A} = \hat{A}^\dagger$$

$$\text{Unitarni operator: } \hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{I}$$

$$\text{Ne.določenosť: } (\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2 \rightarrow \Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

$$\text{Hilbertov prostor } L^2: \forall \psi \in L^2 \exists \text{ štarna baza } \{\psi_n\}, n \in \mathbb{N}_0, \psi \in L^2, \text{ če } \int |\psi|^2 < \infty$$

$$\text{skalarni produkt: } \langle \psi | \psi \rangle = \int_{-\infty}^{\infty} \psi^* \psi dx$$

$$\text{Ket } (|\psi\rangle): \psi \in L^2, \text{ stanje opišemo z vektorjem v } L^2: |\psi\rangle \in L^2$$

$$\text{Bra } (\langle |): \text{Rieszov repr. izrek: } \forall \hat{f} \psi = z \Rightarrow \exists f_z \in L^2, z = \langle f_z | \psi \rangle$$

operator, ki iz funkcije naredi število, je funkcional. Linearne funkcionale

$$\text{pišemo z bra: } \langle f |. \quad \langle f | \psi \rangle = \langle f | z \rangle \in \mathbb{C}$$

3. Operator časovnega razvoja, stacionarna stanja

$$H \neq H(t)$$

$$H|\varphi_n\rangle = E_n|\varphi_n\rangle \rightarrow |\psi(t)\rangle = \sum_n \langle \varphi_n | \psi(0) \rangle e^{-i \frac{E_n}{\hbar} t} |\varphi_n\rangle = \sum_n \langle \varphi_n | \psi(0) \rangle e^{-i \frac{H}{\hbar} t} |\varphi_n\rangle =$$
$$= e^{-i \frac{H}{\hbar} t} \sum_n \langle \varphi_n | \psi(0) \rangle |\varphi_n\rangle = U(t) |\psi(0)\rangle$$

$$\hookrightarrow \text{operator čas. razv. } U(t) = e^{-i \frac{H}{\hbar} t}$$

• ohranja normo val. funkc. : $\langle \psi(t) | \psi(t) \rangle = \langle U(t) \psi(0) | U(t) \psi(0) \rangle = \langle \psi(0) | U^\dagger U | \psi(0) \rangle =$
 $= \langle \psi(0) | \psi(0) \rangle$

Transkacija v času: $U(t_2, t_1) = e^{-i \frac{H}{\hbar} (t_2 - t_1)}$, $\psi(\vec{r}, t_2) = U(t_2, t_1) \psi(\vec{r}, t_1)$

$$|\delta \psi\rangle = |\psi(\vec{r}, t+dt)\rangle - |\psi(\vec{r}, t)\rangle = |U(t+dt, t) \psi(\vec{r}, t)\rangle - |\psi(\vec{r}, t)\rangle$$

$$= \cancel{2} \left(1 - i \frac{H}{\hbar} dt + o(H^2) \right) |\psi(\vec{r}, t)\rangle - |\psi(\vec{r}, t)\rangle = \cancel{2} \left(-i \frac{H}{\hbar} |\psi(\vec{r}, t)\rangle dt \right)$$

$$\Rightarrow \text{itd } \frac{d|\psi\rangle}{dt} = H|\psi(t)\rangle \quad \text{SSE (dobili smo jo samo iz dejstva, da je časovni razvoj unitaren)}$$

4. Simetrije: inverzija, obrat časa, transkacija

-~~inverzija~~ Transkacija

$$\tilde{\psi}(x) = U(s) \psi(x) = \psi(x-s) = \psi(x) - s \frac{\partial}{\partial x} \psi(x) + \frac{s^2}{2!} \frac{\partial^2}{\partial x^2} \psi(x) - \dots =$$

$$= e^{-s \frac{\partial}{\partial x}} \psi(x) = e^{-i s \frac{\hat{p}}{\hbar}} \psi(x)$$

3D: premik za s vzdolž $\vec{n}$ (enotski): $U(s\vec{n}) = e^{-i s \frac{\vec{n} \cdot \hat{\vec{p}}}{\hbar}}$

$$\langle \tilde{x} \rangle = \langle \psi(x-s) | x | \psi(x-s) \rangle = \langle \psi(x) | U^\dagger(s) x U(s) | \psi(x) \rangle = \langle x \rangle + s$$

- inverzija

Operator parnosti: $P: \psi(\vec{r}) \rightarrow \psi(-\vec{r})$

• unitaren: $P P \psi(\vec{r}) = P \psi(-\vec{r}) = \psi(\vec{r}) \Rightarrow P P = I$

• hermitski: $\langle \psi(\vec{r}) | P | \psi(\vec{r}) \rangle = \langle \psi(\vec{r}) | \psi(-\vec{r}) \rangle = \langle \psi(-\vec{r}) | \psi(\vec{r}) \rangle = \langle P \psi(\vec{r}) | \psi(\vec{r}) \rangle$

• $P \nabla \psi = -\nabla P \psi \Rightarrow P \nabla = -\nabla P$; $P \nabla^n = (-1)^n \nabla^n P$

$\rightarrow P p = -p P$ in $P(\vec{r} \times p) = P L = L P \Rightarrow [P, L] = 0$

• če $V(\vec{r}) = V(-\vec{r})$: $[P, H] = 0$

$\Rightarrow$ stanje $|\psi(-\vec{r})\rangle$ je stac., če je $|\psi(\vec{r})\rangle$ stac.: $H|\psi(\vec{r})\rangle = E|\psi(\vec{r})\rangle \Rightarrow H|\psi(-\vec{r})\rangle = E|\psi(-\vec{r})\rangle$

$\rightarrow \psi_{\pm}(\vec{r}) = \frac{1}{\sqrt{2}}(\psi(\vec{r}) \pm \psi(-\vec{r}))$, $P \psi_{\pm}(\vec{r}) = \pm \psi_{\pm}(\vec{r})$ ~~stanje~~ funkcija s sodo in liho parnostjo

- obrat časa ~~čas~~

• boljši izraz: obrat smeri gibanja

$\tau: t \rightarrow -t$

$\psi(\vec{r}, t) = \psi(\vec{r}) e^{-i \frac{E}{\hbar} t} \rightarrow \tau \psi(\vec{r}, -t) = \tau \psi(\vec{r}) e^{i \frac{E}{\hbar} t} = \psi^*(\vec{r}) e^{-i \frac{E}{\hbar} t}$

$\Rightarrow$ za isto last. vr. E sta možni 2 funkciji: $H\psi(\vec{r}) = E\psi(\vec{r})$ in $H\psi^*(\vec{r}) = E\psi^*(\vec{r})$

če spekter ni deg.: $\tau \psi = \psi^* = e^{i\delta} \psi$

$\tau p = -p \tau$, $\tau L = -L \tau$, ampak $\tau H = H \tau$, $x \tau = \tau x$

$\Rightarrow \tau [x, p] = -[p x, p] \tau$. Če želimo, da je $[x, p] = i\hbar$ invariantna na obrat časa, mora veljati: $\tau i = -i \tau \Rightarrow \tau$ antiunitaren.

5. Landauovi nivoji

- ne $\exists \vec{A}$, ki bi bil invarianten na translacije v xy ravnini in rotacije okoli $\hat{z}$ hkrati.

- Landauova umeritev: $\vec{A} = B(-y, 0, 0) \rightarrow \nabla \times \vec{A} = (0, 0, B) \quad \checkmark$

$$\Rightarrow \text{SE: } \frac{1}{2m} \left[(-i\hbar \frac{\partial}{\partial x} + eBy)^2 - \hbar^2 \frac{\partial^2}{\partial y^2} - \hbar^2 \frac{\partial^2}{\partial z^2} \right] \Psi + e\phi \Psi = E \Psi$$

- nastavek: $\Psi = \exp(i(\frac{p_x}{\hbar}x + \frac{p_z}{\hbar}z)) \chi(y)$, predp. $\phi = \phi(y)$

$$\Rightarrow \frac{1}{2m} \left[(p_x + eBy)^2 - \hbar^2 \frac{d^2}{dy^2} \right] \chi(y) + V(y) \chi(y) = E \chi(y), \quad V(y) = e\phi(y) + p_z^2/2m$$

- cikl. frekv. $\omega_c = |e|B/\hbar m$, magn. dolžina $\ell_B = \sqrt{\hbar/|e|B}$, $p_x = \hbar k$, $y_k = -\ell_B^2 k$

$$\Rightarrow \text{harmonski oscilator } -\frac{\hbar^2}{2m} \frac{d^2 \chi}{dy^2} + \frac{1}{2} m \omega_c^2 (y - y_k)^2 \chi + V \chi = E \chi$$

- $V=0$: $\Psi_{nk} = \frac{1}{\sqrt{2\pi}} e^{ikx} \varphi_n(y - y_k)$, $E_n = (n + \frac{1}{2}) \hbar \omega_c \rightarrow$ visoka stopnja degeneracije $k \in \mathbb{R}$

- poenostavljeno: $\Psi(x, 0) = \frac{1}{\sqrt{2\pi}} \int \tilde{\Psi}(k) e^{ikx} dk$, $\Psi(x, t) = \Psi(x, 0) e^{-i \frac{E}{\hbar} t}$

- Lokalne umeritvene transformacije

$$\vec{A}' = \vec{A} + \nabla \Lambda, \quad \phi' = \phi - \frac{\partial \Lambda}{\partial t}$$

• $\Psi' = e^{i\delta} \Psi$, $\delta \in \mathbb{R}$. Vstavimo $\vec{A}'$, ϕ' , Ψ' v SE $\Rightarrow$

$$\oint \Lambda = \frac{\hbar}{e} \oint \vec{A} \cdot d\vec{r}, \quad \text{velja } g' = |\Psi'|^2 = |\Psi|^2 = g$$

Landauovi nivoji so diskretni en. nivoji, po katerih se gibljejo delci v EMP

6. Magnetni moment, Stern-Gerlach, merjenje spina, polarizacija

$$\vec{S} = (S_x, S_y, S_z), \quad [S_\alpha, S_\beta] = i\hbar \epsilon_{\alpha\beta\gamma} S_\gamma, \quad S_\pm = S_x \pm iS_y = S_\mp^\dagger$$

$$S_\pm |s, m\rangle = \hbar \sqrt{s(s+1) - m(m\pm 1)} |s, m\pm 1\rangle$$

Kvantna stanja so spinorji, oznake: $|s, m\rangle = |s, m\rangle = |\frac{1}{2}, m\rangle = |m\rangle = \begin{cases} |\uparrow\rangle \\ |\downarrow\rangle \end{cases}$

$$S^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle, \quad S_z |s, m\rangle = \hbar m |s, m\rangle, \quad S_+ |\downarrow\rangle = \hbar |\uparrow\rangle, \quad S_- |\uparrow\rangle = \hbar |\downarrow\rangle$$

$$S_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = S_-^\dagger, \quad S_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \rightarrow S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$S_\alpha^2 = \frac{\hbar^2}{4} I, \quad [S_\alpha, S_\beta^2] = 0, \quad [S_\alpha^2, S_\beta^2] = 0$$

Magnetni moment

$$\mu_B = \frac{e\hbar}{2m_e} \text{ Bohrov magneton}, \quad \hat{\mu}_L = -\frac{\mu_B}{\hbar} \hat{L} \text{ tirni mag. moment}$$

$$\hat{\mu}_S = -g_s \frac{\mu_B}{\hbar} \hat{S} \text{ spinski mag. moment} \quad \mu_{L,z} = -m_L \mu_B, \quad \mu_{S,z} = \mp \frac{1}{2} g_s \mu_B$$

$\hookrightarrow \text{za } e^-$

Stern-Gerlach

$$\vec{H} = \vec{\mu} \times \vec{B}_0, \quad \vec{B}_0 = (0, 0, B_z)$$


magnet

$$\chi(0) = \begin{pmatrix} \cos \frac{\varphi}{2} \\ \sin \frac{\varphi}{2} \end{pmatrix}$$

$\hookrightarrow$ spinor

φ ... smer v ravnini xz

$$\vec{n}_0 = \langle \sigma \rangle = (\sin \varphi, 0, \cos \varphi) \text{ smer dip. momenta}$$

Po preletu magnetu ($t = \tau$): $\chi(\tau) = \begin{pmatrix} e^{i\delta\varphi/2} \cos \frac{\varphi}{2} \\ e^{-i\delta\varphi/2} \sin \frac{\varphi}{2} \end{pmatrix}, \quad \delta\varphi = \omega_s \tau = \omega_s \frac{L}{v_y}$

$$\vec{n}(\tau) = \langle \chi(\tau) | \sigma | \chi(\tau) \rangle = (\sin \varphi \cos \delta\varphi, -\sin \varphi \sin \delta\varphi, \cos \varphi)$$

Ker ima magnet nehomogeno polje: $\vec{B} = (B_x, 0, B_z)$:

$$\vec{F} = \nabla(\vec{\mu} \cdot \vec{B}), \text{ razvijemo } \vec{B} = \vec{B}_0 + \sum_\alpha \frac{\partial \vec{B}}{\partial x_\alpha} x_\alpha + \dots$$

$$\vec{F} \approx F_z \hat{e}_z = \mu \frac{\partial B_z}{\partial z} \Big|_{z=0} \hat{e}_z \rightarrow m v_z = \pm F_z \tau = \pm F_z \frac{L}{v_y}, \quad \text{ ~~$\langle \sigma_z \rangle = \cos \varphi$~~$$

$$P_\uparrow = \cos^2 \frac{\varphi}{2}, \quad P_\downarrow = \sin^2 \frac{\varphi}{2}, \quad \text{ ~~$\langle \sigma_z \rangle = \cos \varphi$~~$$

$$\langle \sigma_z \rangle = \langle \chi | \sigma_z | \chi \rangle = \begin{pmatrix} \cos \frac{\varphi}{2} & \sin \frac{\varphi}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos \frac{\varphi}{2} \\ \sin \frac{\varphi}{2} \end{pmatrix} = \dots = \cos \varphi$$

Stern-Gerlach (kvantno)

$$H = \underbrace{-\frac{\hbar^2}{2m} \nabla^2}_{H_0} - \frac{e}{m_e} \vec{S} \cdot \vec{B} = H_0 - E_z \sigma_z - F z \sigma_z, \quad 2E_0 = \frac{e\hbar}{m_e} B_0, \quad F = \frac{e\hbar}{2m_e} \left. \frac{\partial B_z}{\partial z} \right|_{z=0}$$

pred magnetom: $\Psi(\vec{r}, 0) = \Psi_0(x, y, 0) \phi_0(z) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$, $\phi_0(z) = \int \tilde{\phi}(p) e^{i \frac{p}{\hbar} z} dp$,

$$|\chi(0)\rangle = \begin{pmatrix} \cos \frac{\varphi}{2} \\ \sin \frac{\varphi}{2} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \quad v_y = \frac{1}{m} \langle p_y \rangle \dots \text{velika} \Rightarrow \Psi(\vec{r}, t) = \Psi_0(x, y, t) \begin{pmatrix} \phi_+(z, t) \\ \phi_-(z, t) \end{pmatrix}$$

$$\rightarrow SE: i\hbar \frac{\partial}{\partial t} \begin{pmatrix} \phi_+ \\ \phi_- \end{pmatrix} = - \left(\frac{\hbar^2}{2m} \frac{\partial^2}{\partial z^2} + \frac{1}{2} \mu_B \sigma_z + F z \sigma_z \right) \begin{pmatrix} \phi_+ \\ \phi_- \end{pmatrix}$$

po magnetu: $\Psi(\vec{r}, t) = e^{-i \frac{H_0}{\hbar} t} \Psi(\vec{r}, 0)$

$$\begin{pmatrix} \phi_+(z, t) \\ \phi_-(z, t) \end{pmatrix} = \int \begin{pmatrix} e^{i \frac{1}{2} \sigma \varphi} c_1 \tilde{\phi}(p - \delta p) \\ e^{-i \frac{1}{2} \sigma \varphi} c_2 \tilde{\phi}(p + \delta p) \end{pmatrix} e^{i \frac{p}{\hbar} z} dp$$

$$\rightarrow \Psi(\vec{r}, t) = \Psi_0(x, y, t) \begin{pmatrix} e^{i \frac{1}{2} \sigma \varphi} c_1 \phi_0(z - vt) e^{i \frac{\delta p}{\hbar} (z - \frac{1}{2} vt)} \\ e^{-i \frac{1}{2} \sigma \varphi} c_2 \phi_0(z + vt) e^{-i \frac{\delta p}{\hbar} (z + \frac{1}{2} vt)} \end{pmatrix}$$

7. Teorija motenj, 1. nedegeneriran ~~red~~ red

$$H = H_0 + H_1, \quad H_0 |n^0\rangle = E_n^{(0)} |n^0\rangle, \quad \langle m^0 | n^0 \rangle = \delta_{mn} \quad (\text{poznano last. st. } H_0)$$

$\hookrightarrow$ majhen

$$H_1 = \lambda V, \quad \lambda \text{ pomalen brezdim. par., po katerem razvijamo}$$

$$H |n\rangle = E_n |n\rangle \rightarrow |n\rangle = |n^0\rangle + \lambda |n^1\rangle + \dots + \lambda^m |n^m\rangle + \dots \quad (1)$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \dots$$

$$(1) / \langle n^0 | : \quad \langle n^0 | n \rangle = 1 + \lambda \langle n^0 | n^1 \rangle + \lambda^2 \langle n^0 | n^2 \rangle + \dots = 1 \quad (\text{postavimo})$$

$$\rightarrow SE: (H_0 + \lambda V) (|n^0\rangle + \lambda |n^1\rangle + \dots) = (E_n^{(0)} + \lambda E_n^{(1)} + \dots) (|n^0\rangle + \lambda |n^1\rangle + \dots)$$

velja $\forall \lambda \Rightarrow \lambda^0: H^0 |n^0\rangle = E_n^{(0)} |n^0\rangle, \quad \lambda^1: H_0 |n^1\rangle + V |n^0\rangle = E_n^{(0)} |n^1\rangle + E_n^{(1)} |n^0\rangle$

$$\rightarrow \cancel{E_n^{(0)} \langle n^0 | n^1 \rangle} + \langle n^0 | V | n^0 \rangle = \cancel{E_n^{(0)} \langle n^0 | n^1 \rangle} + E_n^{(1)} \underbrace{\langle n^0 | n^0 \rangle}_1$$

$$\Rightarrow E_n^{(1)} = \langle n^0 | V | n^0 \rangle = V_{nn}$$

8. Kontinuitetna enačba, lastnosti valovne funkcije

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{j} = 0$$

$$\frac{\partial}{\partial t} |\Psi|^2 = \frac{\partial \Psi^*}{\partial t} \Psi + \Psi^* \frac{\partial \Psi}{\partial t} = \dots = - \frac{\partial}{\partial x} \left(\frac{\hbar}{2im} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \right) - \frac{2}{\hbar} \ln(V) |\Psi|^2$$

$\underbrace{\hspace{10em}}_{\vec{j}} \quad \underbrace{\hspace{10em}}_q$

* q poravnadi 0, ker $V \in \mathbb{R}$

- Normalizacija: $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1 \quad \forall \Psi \in \mathbb{C}$

- poravnadi Ψ iz Schwartzovega prostora (te funkcije, ki v $\pm\infty$ padajo hitreje kot kak polinom): $\int_{-\infty}^{\infty} x^n |\Psi|^2 dx < \infty \quad \forall n$

- Pričakovana vrednost:

$$\langle x \rangle = \int \Psi^* x \Psi dx$$

- Hitrost težišča porazd.: $v_T = \frac{d\langle x \rangle}{dt} = \int \left(\frac{\partial \Psi^*}{\partial t} x \Psi + \Psi^* \frac{\partial x}{\partial t} \Psi + \Psi^* x \frac{\partial \Psi}{\partial t} \right) dx$

$$= \dots = \frac{\hbar^2}{2im} \int \frac{\partial}{\partial x} \left(\frac{\partial \Psi^*}{\partial x} x \Psi - \Psi^* x \frac{\partial \Psi}{\partial x} \right) dx + \frac{1}{m} \int \Psi^* \underbrace{\left(-i\hbar \frac{\partial}{\partial x} \right)}_{\hat{p}} \Psi dx$$

$$\Rightarrow \hat{p} = -i\hbar \frac{\partial}{\partial x} \text{ oz. } \hat{p} = -i\hbar \nabla$$

9. Hermitski operatorji, projektorji, opazljivke, baza

Hermitski operator: $\langle \Psi | A | \Psi \rangle = \langle \Psi | A \Psi \rangle = \langle A \Psi | \Psi \rangle$
(simetričen)

• $\langle A \rangle \in \mathbb{R}$: $\langle a | A | a \rangle = \langle A | a \rangle = \langle a | A a \rangle^*$

• $\langle \Psi | A^2 | \Psi \rangle = \langle A \Psi | A \Psi \rangle \geq 0$

• Lastne vrednost. $\in \mathbb{R}$: $A|a\rangle = a|a\rangle \quad / \cdot \langle a| \Rightarrow \langle a | A | a \rangle = a \langle a | a \rangle \in \mathbb{R}$

• $A|a\rangle = a|a\rangle, A|b\rangle = b|b\rangle \rightarrow \langle b | A | a \rangle = a \langle b | a \rangle, \quad \mathbb{R} \ni a \in \mathbb{R}$

$$\langle a | A | b \rangle = b \langle a | b \rangle \quad / ^*$$

$$\rightarrow \langle a | A | b \rangle^* = b \langle a | b \rangle^* \rightarrow \langle A a | b \rangle^* = \langle b | A a \rangle = b \langle b | a \rangle$$

Projektor: $|a\rangle\langle a|$

$$0 = (a-b) \langle b | a \rangle \Rightarrow \langle b | a \rangle = 0 \text{ če } b \neq a \Rightarrow |a\rangle \perp |b\rangle$$

• Opazljivke: vsaki merilni količini A priredimo operator $\hat{A}$, ki deluje na stanja v Hilb. prost. $\hat{A}$ mora biti hermitski

• Lastna stanja $\{|a_i\rangle\}$; opazljivke $\hat{A}$ tvorijo ONB:
 $\langle a_i | a_j \rangle = \delta_{ij}$, $\sum_i |a_i\rangle \langle a_i| = I$

→ $|\psi\rangle$ lahko ~~zapišemo~~ razvijemo po bazi:

$$|\psi\rangle = I|\psi\rangle = \sum_i |a_i\rangle \langle a_i | \psi \rangle = \sum_i c_i |a_i\rangle$$

10. Reprezentacija $|p\rangle$, $|x\rangle$, $\langle x|\psi\rangle$, nepravi vektorji

- koordinatna reprezentacija

$$\langle x | x' \rangle = \delta(x - x'), \quad \int_{-\infty}^{\infty} |x\rangle \langle x| dx = I, \quad \psi(x) = \langle x | \psi \rangle$$

operatorji: $\hat{x} = x$, $\hat{p} = -i\hbar \frac{\partial}{\partial x}$

- gibalna reprezentacija

$$\langle p | p' \rangle = \delta(p - p'), \quad \int_{-\infty}^{\infty} |p\rangle \langle p| dp = I, \quad \phi(p) = \langle p | \psi \rangle$$

operatorji: $\hat{p} = p$, $\hat{x} = i\hbar \frac{\partial}{\partial p}$

- prehod med bazama (FT): $\langle x | p \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{px}{\hbar}}$

$$x \rightarrow p: \phi(p) = \langle p | \psi \rangle = \int_{-\infty}^{\infty} \underbrace{\langle p | x \rangle}_{\frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{px}{\hbar}}} \underbrace{\langle x | \psi \rangle}_{\psi(x)} dx = \frac{1}{\sqrt{2\pi\hbar}} \int e^{i\frac{px}{\hbar}} \psi(x) dx$$

$$p \rightarrow x: \psi(x) = \langle x | \psi \rangle = \int_{-\infty}^{\infty} \underbrace{\langle x | p \rangle}_{\frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{px}{\hbar}}} \underbrace{\langle p | \psi \rangle}_{\phi(p)} dp = \frac{1}{\sqrt{2\pi\hbar}} \int e^{i\frac{px}{\hbar}} \phi(p) dp$$

Nepravi vektorji (npr. $\hat{x}$, $\hat{p}$): uvedel jih je Dirac, da je omogočil matematični opis kv. meh. Značilnosti: ne-normalizabilnost, povezava z Diracovo δ : normalni ^{bazni} vektorji $\langle m | n \rangle = \delta_{mn}$, nepravi bazni vektorji: $\langle p_1 | p_2 \rangle = \delta(p_1 - p_2)$

11. Centralni potencial

$$\cdot V(\vec{r}) = V(r) \Rightarrow [L^2, H] = [L_z, H] = 0$$

$$\Psi(\vec{r}) = R(r) Y_l^m(\vartheta, \varphi) = \frac{u(r)}{r} Y_l^m(\vartheta, \varphi)$$

$$\rightarrow SE: -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + V_{eff}(r) u(r) = E u(r), \quad V_{eff} = V(r) + \frac{l(l+1)\hbar^2}{2mr^2}$$

$$\cdot r \rightarrow 0: E, V(r) \text{ zanemarimo} \rightarrow \frac{d^2 u}{dr^2} = \frac{l(l+1)}{r^2} u(r) \rightarrow u(r) = C_1 r^{l+1} + D_1 r^{-l}$$

$$D_1 = 0 \text{ (normalizacija) za } l > 0$$

$$\cdot l = 0: \text{ ~~R(r)~~ } \Psi(r) = C_0 + \frac{D_0}{r}$$

Poisson za točkast naboj: za odvisnost
rešitve $\frac{1}{r}$ potrebujemo v središčnem
potencial $V \propto \delta^3(\vec{r}) \Rightarrow D_0 = 0$

$$\rightarrow \underline{R(r) = C_1 r^l}$$

$$\cdot r \rightarrow \infty, V(r \rightarrow \infty) \rightarrow 0$$

$$1) E > 0 \text{ (kontinuumski stanja): } -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} = E u(r) \rightarrow u(r) = C_+ e^{ikr} + C_- e^{-ikr}$$

$$2) E < 0 \text{ (diskretna stanja): } u(r) = D_- e^{-kr} + \text{~~0~~} D_+ e^{+kr}, \quad k = \sqrt{\frac{2mE}{\hbar^2}}, \quad K = \sqrt{\frac{2m|E|}{\hbar^2}}$$

Pri $E = E_n$ velja $D_+ = 0$ in $u_n(r)$ vezano stanje brez degeneracije

Za diskretna stanja mora veljati $\lim_{r \rightarrow \infty} r V(r) = 0$

12. Umeritvene transformacije in vpliv na $|\Psi\rangle$

1) Globalna: če Ψ pomnožimo z $e^{i\delta}$, to ne vpliva na rezultate

$$|\tilde{\Psi}\rangle = e^{i\delta} |\Psi\rangle \rightarrow \langle \tilde{\Psi} | \tilde{A} | \tilde{\Psi} \rangle = \langle \Psi | A | \Psi \rangle$$

2) Lokalna: $\tilde{\Psi}(\vec{r}, t) = e^{i\delta(\vec{r}, t)} \Psi(\vec{r}, t) \rightarrow |\tilde{\Psi}|^2 = |\Psi|^2$, ampak spremenijo SE,
vzamejo še transformiramo tudi EM potenciale:

$$\vec{A}' = \vec{A} + \nabla \Lambda, \quad \phi' = \phi - \frac{\partial \Lambda}{\partial t} \rightarrow i\hbar \frac{\partial}{\partial t} \Psi' = \frac{1}{2m} (i\hbar \nabla - e\vec{A}')^2 \Psi' + e\phi' \Psi'$$

$$\dots \Rightarrow i\hbar \frac{\partial}{\partial t} \Psi = \frac{1}{2m} \sum_{\alpha} \left(-i\hbar \frac{\partial}{\partial x_{\alpha}} - eA'_{\alpha} + \hbar \frac{\partial \delta}{\partial x_{\alpha}} \right)^2 \Psi + (e\phi' + \hbar \frac{\partial \delta}{\partial t}) \Psi$$

$$\text{vzamemo } \Lambda = \frac{\hbar}{e\delta} \Rightarrow \checkmark$$

-Vplivi na opazljivke: opazljivke so invariantne na umeritev, $\vec{p}$ ni invariant
ampak $\vec{\pi} = \vec{p} - e\vec{A}$ (kinetični impulz) je. Pri izračunu verjetnostnega toka
p moramo namesto $\vec{p}$ uporabiti $\vec{\pi}$, da je invarianten.

13. Teorija motenj, udelež. 2. red

$$H = H_0 + H_1, \quad H_0 |n^0\rangle = E_n^{(0)} |n^0\rangle, \quad \langle m^0 | n^0 \rangle = \delta_{mn} \text{ (poznamo last. st. } H_0)$$

Wannier

$H_1 = \lambda V$, λ pomožni brezdim. par., po katerem razvijamo

$$H |n\rangle = E_n |n\rangle \Rightarrow |n\rangle = |n^0\rangle + \lambda |n^1\rangle + \lambda^2 |n^2\rangle + \dots \quad (1)$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \dots$$

(1) / $\langle n^0 | \Rightarrow \langle n^0 | n \rangle = 1 + \lambda \langle n^0 | n^1 \rangle + \lambda^2 \langle n^0 | n^2 \rangle + \dots = 1$ (postavimo)

SE: $(H_0 + \lambda V)(|n^0\rangle + \lambda |n^1\rangle + \dots) = (E_n^{(0)} + \lambda E_n^{(1)} + \dots)(|n^0\rangle + \lambda |n^1\rangle + \dots) + \lambda$

$\Rightarrow \lambda^0: H^0 |n^0\rangle = E_n^0 |n^0\rangle \quad \checkmark$ (trivialno)

$\lambda^2: H_0 |n^2\rangle + V |n^1\rangle = E_n^{(0)} |n^2\rangle + E_n^{(1)} |n^1\rangle + E_n^{(2)} |n^0\rangle \quad / \langle n^0 |$

$$E_n^{(0)} \underbrace{\langle n^0 | n^2 \rangle}_0 + \langle n^0 | V | n^1 \rangle = E_n^{(0)} \cancel{\langle n^0 | n^2 \rangle} + E_n^{(1)} \cancel{\langle n^0 | n^1 \rangle} + E_n^{(2)} \underbrace{\langle n^0 | n^0 \rangle}_1$$

$\Rightarrow E_n^{(2)} = \langle n^0 | V | n^1 \rangle$

$|n^j\rangle = I |n^j\rangle = \sum_{m \neq n} |m^0\rangle \langle m^0 | n^j \rangle$

$\lambda^1: H_0 |n^1\rangle + V |n^0\rangle = E_n^{(0)} |n^1\rangle + E_n^{(1)} |n^0\rangle \quad / \sum \langle m^0 |$

$\Rightarrow (E_n^{(0)} - E_m^{(0)}) \langle m^0 | n^1 \rangle = \langle m^0 | V | n^0 \rangle = V_{mn}$

$\Rightarrow |n^1\rangle = \sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} |m^0\rangle$

$\Rightarrow E_n^{(2)} = \underbrace{\langle n^0 | V}_{\sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}} \underbrace{\sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} |m^0\rangle}_{= \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}}$

14. $\frac{d\langle x \rangle}{dt}$, operator p , $\frac{d\langle A \rangle}{dt}$

$$\psi(t) = e^{-iHt/\hbar} \psi(0)$$

$$A(t) = U^\dagger A U$$

$$\langle A \rangle = \langle \psi | A | \psi \rangle$$

$$\frac{d\langle A \rangle}{dt} = \frac{d\langle \psi |}{dt} A | \psi \rangle + \langle \psi | \frac{\partial A}{\partial t} | \psi \rangle + \langle \psi | A \frac{\partial | \psi \rangle}{\partial t}$$

se: $\frac{1}{i\hbar} H | \psi \rangle$

$$\Rightarrow \frac{d\langle A \rangle}{dt} = \left\langle \frac{\partial A}{\partial t} \right\rangle - \frac{1}{i\hbar} \langle \psi | H A | \psi \rangle + \frac{1}{i\hbar} \langle \psi | A H | \psi \rangle =$$

$$= \left\langle \frac{\partial A}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle \psi | A H - H A | \psi \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle$$

$$[x, H] = [x, \frac{p^2}{2m}] + [x, V(x)] = \frac{1}{2m} (p[x, p] + [x, p]p) = \frac{i\hbar}{2m} p(1+1) = \frac{i\hbar}{m} p$$

$$\left\langle \frac{\partial x}{\partial t} \right\rangle = 0 \Rightarrow \frac{d\langle x \rangle}{dt} = \frac{p}{m}$$

$$[p, H] = [p, \frac{p^2}{2m}] + [p, V(x)] \quad [p, V] \psi = -i\hbar \left(\frac{\partial}{\partial x} (V\psi) - V \frac{\partial \psi}{\partial x} \right) = -i\hbar \frac{\partial V}{\partial x}$$

$$\frac{d\langle p \rangle}{dt} = \left\langle \frac{\partial p}{\partial t} \right\rangle + \frac{1}{i\hbar} \langle [p, H] \rangle = - \left\langle \frac{\partial V}{\partial x} \right\rangle = \langle F(x) \rangle$$

15. Matrični zapis operatorja, unitarni operatorji, razvoj vektorja po bazi

baza $\{|n\rangle\} \rightarrow A_{mn} = \langle m | A | n \rangle$, v splošnem ∞ -dim.

$$|\psi\rangle = \sum c_n |n\rangle \Rightarrow A|\psi\rangle = |\psi_1\rangle = \sum d_n |n\rangle \text{ se prevede na } \begin{pmatrix} d_1 \\ \vdots \end{pmatrix} = \begin{bmatrix} A_{11} & A_{12} & \dots \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{pmatrix} c_1 \\ \vdots \end{pmatrix}$$

Primeri: $a, a^\dagger$ imata matr. elemente samo nad/pod diagonalo

L_2, L_3, L_z so bločno diagonalne glede na l

spin-Paulijevе matrice

razvoj vekt. po bazi: $|\psi\rangle = \sum c_n |n\rangle$, $c_n = \langle n | \psi \rangle$

Unitarni operatorji: $UU^\dagger = U^\dagger U = I$

$$\langle \psi | \psi \rangle = \langle U\psi | U\psi \rangle$$

$$\langle \psi | A | \psi \rangle = \langle U^\dagger U\psi | A | U^\dagger U\psi \rangle = \langle U\psi | U A U^\dagger | U\psi \rangle$$

$$|\tilde{\psi}\rangle = U|\psi\rangle \Rightarrow \langle \tilde{\psi} | = \langle U\psi | = \langle \psi | U^\dagger$$

$$\{|\psi_n\rangle\} \rightarrow \{|\tilde{\psi}_n\rangle\}: U = U \sum_n |\psi_n\rangle \langle \psi_n| = \sum_n |\tilde{\psi}_n\rangle \langle \psi_n| = \sum_{mn} |\psi_m\rangle \underbrace{\langle \psi_m | \tilde{\psi}_n \rangle}_{U_{mn}} \langle \psi_n|$$

$$\rightarrow UU^\dagger = \sum_n |\tilde{\psi}_n\rangle \langle \tilde{\psi}_n|$$

$$\{|\psi_n\rangle\} \rightarrow \{|\tilde{\psi}_n\rangle\}: |\psi\rangle = \sum_n c_n |\psi_n\rangle = \sum_{mn} |\tilde{\psi}_m\rangle \langle \tilde{\psi}_m | c_n |\psi_n\rangle = \sum_n d_n |\tilde{\psi}_n\rangle, d_n = \sum_m U_{nm}^\dagger c_m$$

- $A|a\rangle = a|a\rangle \rightarrow UAU^\dagger U|a\rangle = aU|a\rangle \rightarrow \tilde{A}|\tilde{a}\rangle = a|\tilde{a}\rangle$
 - če K hermitski, je $U = e^{iK}$ unitaren, saj $UU^\dagger = e^{iK}e^{-iK} = I$
 - vsak enoparametričen unitaren operator se da zapisati kot $U(s) = e^{isK}$, $K = K^\dagger$
↓
 generator unitarnega operatorja
16. Harmonijski oscilator: a, x, p, H v matrični obliki

$$a|n\rangle = \sqrt{n}|n-1\rangle \rightarrow a_{mn} = \langle m|a|n\rangle = \sqrt{n}\langle m|n-1\rangle = \sqrt{n}\delta_{m,n-1}$$

$$a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \rightarrow a_{mn}^\dagger = \langle m|a^\dagger|n\rangle = \sqrt{n+1}\delta_{m,n+1}$$

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a+a^\dagger) \quad p = i\sqrt{\frac{m\hbar\omega}{2}}(a^\dagger - a) \quad H = \hbar\omega(a^\dagger a + \frac{1}{2}I) \rightarrow$$

$$H_{mn} = \langle m|H|n\rangle = \hbar\omega(n + \frac{1}{2})\langle m|n\rangle = \hbar\omega(n + \frac{1}{2})\delta_{mn}$$

a : 1 vrsto nad diagonalo in a člene $1, \sqrt{2}, \sqrt{3}, \dots$

$a^\dagger$: -11- pod -11-

x : -11- nad in pod -11-, pomnoženo z $\sqrt{\frac{\hbar}{2m\omega}}$

p : 1 vrsto nad diagonalo $-1, -\sqrt{2}, -\sqrt{3}, \dots$; 1 vrsto pod diag. $1, \sqrt{2}, \sqrt{3}, \dots$ *

H : diagonalna, členi $(n + \frac{1}{2})$ začeni z $n=0$; pomnoženo z $\hbar\omega$

x pomnoženo z $i\sqrt{\frac{m\hbar\omega}{2}}$

17. Coulombski potencial, $E < 0$, degeneracija, kl. limita

$$\left(-\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{l(l+1)\hbar^2}{2mr^2} - \frac{e_0^2}{4\pi\epsilon_0 r}\right)u(r) = Eu(r), \quad \rho = Kr$$

$$\rightarrow \left(-\frac{d^2}{d\rho^2} + \frac{l(l+1)}{\rho^2} - \frac{m e_0^2}{2\pi\epsilon_0 \hbar^2 K} \frac{1}{\rho}\right)u(\rho) = -u(\rho), \quad u(\rho) = \rho^{l+1} v(\rho) e^{-\rho}$$

$$\rightarrow \rho v'' + 2(l+1-\rho)v' + [\rho_0 - 2(l+1)]v = 0, \quad v(\rho) = \sum_{k=0}^{\infty} c_k \rho^k$$

$$\Rightarrow c_{k+1} = \frac{2(k+1) - \rho_0}{(k+1)(k+2)} c_k$$

asimptotsko obnašanje: $v(r) \propto e^{2\beta} \rightarrow u(r) = r^{l+1} e^{2\beta} e^{-\beta} \alpha e^{\beta}$

$\Rightarrow$ ker želimo, da ~~funkc.~~ val. funke. ne poživja v ∞ , se mora vrsta za $v(r)$

končati pri k_{maks} : $c_k = 0$ za $k > k_{maks}$

$$E = \frac{\hbar^2 k^2}{2m} = -\frac{m e^4}{8\pi^2 \epsilon_0^2 \hbar^2} \rightarrow E_n = -\underbrace{\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2}_{R_y} \frac{1}{n^2} \quad (\text{Bohr})$$

Degeneracija: za E_n $\exists$ n razl. vrednosti $l = 0, 1, 2, \dots, n-1$; dodatno še

$(2l+1)$ -degeneracija zaradi $m \Rightarrow E_n$ je n^2 -krat degenerirana,

$$\text{ker } \sum_{l=0}^{n-1} (2l+1) = n^2$$

Klasična limita: ^{pri} $l = n-1$, se Δ obnaša kot Poissonova porazd.

$$\langle r \rangle = n(n+1)a_0, \quad \langle r^2 \rangle = n^2(n+1)\left(n + \frac{1}{2}\right)a_0^2 \rightarrow \Delta r = \frac{1}{2} n \sqrt{2n+1} a_0$$

$$\rightarrow \frac{\Delta r}{\langle r \rangle} = \frac{1}{\sqrt{2n+1}} \rightarrow 0 \Rightarrow \text{Poissonova porazd. preide v Gaussovo.}$$

1.8. Spin, Paulijeve matrice

$$\vec{S} = (S_x, S_y, S_z) \quad [S_\alpha, S_\beta] = i\hbar \sum_{\gamma} \epsilon_{\alpha\beta\gamma} S_\gamma \quad S_\pm = S_x \pm iS_y = S_\mp^\dagger$$

$$S_\pm |s, m\rangle = \hbar \sqrt{s(s+1) - m(m\pm 1)} |s, m\pm 1\rangle$$

$$\text{Kvantna stanja so spinorji} \quad S^2 |s, m\rangle = \hbar^2 s(s+1) |s, m\rangle$$

$$S_z |s, m\rangle = \hbar m |s, m\rangle, \quad S_- |\downarrow\rangle = \hbar |\uparrow\rangle, \quad S_- |\uparrow\rangle = \hbar |\downarrow\rangle$$

$$\rightarrow S_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad S_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \rightarrow \underbrace{S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{\sigma_x}, \quad \underbrace{S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}}_{\sigma_y}, \quad \underbrace{S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}_{\sigma_z}$$

$$\cdot \det \sigma_\alpha = -1$$

$$\cdot \text{tr } \sigma_\alpha = 0$$

$$\cdot \sigma_\alpha^2 = I$$

$$\cdot \sigma_x \sigma_y \sigma_z = iI$$

$$\cdot \sigma_\alpha \sigma_\beta = \delta_{\alpha\beta} I + i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \sigma_\gamma$$

$$\cdot [\sigma_\alpha, \sigma_\beta] = 2i \sum_{\gamma} \epsilon_{\alpha\beta\gamma} \sigma_\gamma$$

$$\cdot \{\sigma_\alpha, \sigma_\beta\} = 2\delta_{\alpha\beta} I$$

Paulijeve matrice

$$\cdot \vec{a} \cdot \vec{\sigma} = \sum_{\alpha} a_{\alpha} \sigma_{\alpha} = (\vec{a} \cdot \vec{\sigma})^\dagger, \quad a_{\alpha} \in \mathbb{R}$$

$$\cdot (\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = \sum_{\alpha\beta} a_{\alpha} \sigma_{\alpha} b_{\beta} \sigma_{\beta} = \vec{a} \cdot \vec{b} I + i \sum_{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} a_{\alpha} b_{\beta} \sigma_{\gamma}$$

19. Clebsch-Gordanovi koeficienti, spin in koordinate

$$[J_{1\alpha}, J_{2\beta}] = i\hbar \delta_{\alpha\beta} \sum_{\gamma} J_{1\gamma}$$

$$|j_1 j_2 j m\rangle = \sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} |j_1 m_1 j_2 m_2\rangle \underbrace{\langle j_1 m_1 j_2 m_2 | j m \rangle}_{C_{j_1 m_1 j_2 m_2}^{j m} \in \mathbb{C}}, \quad m = m_1 + m_2$$

Clebsch-Gordanovi koef.

→ koef. so nenizeljni le ko je izpolnjena trikotniška neenakost:

$$|j_1 - j_2| \leq j \leq j_1 + j_2$$

Tabela:

$j_1 \times j_2$	j
$m_1 \quad m_2$	C.G.

* C.-G. je treba koreniti in pomnožiti s predznakom: $-\frac{1}{2} \rightarrow -\sqrt{\frac{1}{2}}$

Tabela se uporablja za prehod med bazami

Zgled: izračun stanj (in $\frac{1}{2}$ $J=L+S$, $j_1=L$, $j_2=S$
 $\{ |L \frac{1}{2} j m\rangle \}$ $\{ |L m_1\rangle \}$ $\{ |S \uparrow, \downarrow\rangle \}$

• $l(=0)$: $j=\frac{1}{2}$: $|0 \frac{1}{2} \frac{1}{2} \pm \frac{1}{2}\rangle = |00\rangle \begin{matrix} \uparrow\uparrow \\ \downarrow\downarrow \end{matrix}$

• $p(l=1)$: 1) $j=\frac{3}{2}$: $\rightarrow m=\frac{3}{2} \rightarrow |1 \frac{1}{2} \frac{3}{2} \frac{3}{2}\rangle = |11\rangle \uparrow\uparrow$

$$J_+ = L_+ + S_+, \quad J_- |j m\rangle = \hbar \sqrt{j(j+1) - m(m-1)} |j, m-1\rangle$$

$$J_- | \frac{3}{2} \frac{3}{2} \rangle = \hbar \sqrt{3} | \frac{3}{2} \frac{1}{2} \rangle, \quad (L_- + S_-) | 1, \frac{1}{2} \rangle = \hbar \sqrt{2} | 0 \frac{1}{2} \rangle + \hbar | 1 - \frac{1}{2} \rangle$$

$$\rightarrow | \frac{3}{2} \frac{1}{2} \rangle = \sqrt{\frac{2}{3}} | 0 \uparrow \rangle + \frac{1}{\sqrt{3}} | 1 \downarrow \rangle, \quad \text{ponovimo še } 2 \times (m = -\frac{1}{2}, m = -\frac{3}{2})$$

2) $j=\frac{1}{2}$: $m=\frac{1}{2}$ mora biti $\perp$ na $| \frac{3}{2} \frac{1}{2} \rangle \Rightarrow | \frac{1}{2} \frac{1}{2} \rangle = \sqrt{\frac{2}{3}} | 1 - \frac{1}{2} \rangle - \frac{1}{\sqrt{3}} | 0 \frac{1}{2} \rangle$
 $m=-\frac{1}{2}$: uporabimo J_-

20. $H = H(t)$, 1. red časovno odvisne perturbacije

$$H(t) = H_0 + \lambda V(t), \quad H_0 |n\rangle = E_n |n\rangle, \quad \langle m | n \rangle = \delta_{mn}$$

Razvijemo zač. stanje: $|\psi(0)\rangle = \sum_n c_n(0) |n\rangle \rightarrow |\psi(t)\rangle = \sum_n c_n(t) e^{-i \frac{E_n}{\hbar} t} |n\rangle$

$$\rightarrow \text{SE: } i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H |\psi(t)\rangle$$

$$i\hbar \sum_n \left(\frac{\partial c_n(t)}{\partial t} e^{-i \frac{E_n}{\hbar} t} - i \frac{E_n}{\hbar} c_n(t) e^{-i \frac{E_n}{\hbar} t} \right) |n\rangle = \sum_n (E_n + \lambda V(t)) c_n(t) e^{-i \frac{E_n}{\hbar} t} |n\rangle / \langle m |$$

$$i\hbar \frac{\partial c_m(t)}{\partial t} e^{-i \frac{E_m}{\hbar} t} + \cancel{E_m c_m(t) e^{-i \frac{E_m}{\hbar} t}} = \cancel{E_m c_m(t) e^{-i \frac{E_m}{\hbar} t}} + \lambda \sum_n \langle m | V(t) | n \rangle e^{-i \frac{E_n}{\hbar} t} c_n(t)$$

$$V_{mn}(t) = \langle m | V(t) | n \rangle \exp\left(-i \frac{E_n - E_m}{\hbar} t\right)$$

$$\rightarrow \underline{i\hbar \frac{\partial}{\partial t} |c(t)\rangle = \lambda V(t) |c(t)\rangle} \quad \text{Matrična enačba}$$

Predp., da je motnja šibka in $V(t) = \begin{cases} 0, & t < 0 \\ V(t), & t \geq 0 \end{cases}$ ter $|\psi(0)\rangle = |m\rangle$,

oziroma $c_n^{(0)}(0) = \delta_{mn} \rightarrow i\hbar \frac{\partial c_k^{(1)}(t)}{\partial t} = \sum_n V_{kn}(t) c_n^{(0)}(t) = V_{km}(t)$

(uporabili smo približek $c_n^{(0)}(t) = \delta_{mn}$ za $t > 0$ (koeficienti ~~se~~ v prvem redu se spremenijo)

$$\rightarrow \underline{c_k^{(1)}(t) = \frac{1}{i\hbar} \int_0^t V_{km}(t') dt'} \rightarrow |\psi(t)\rangle = \sum_k \left(\delta_{km} + \lambda c_k^{(1)}(t) + O(\lambda^2) \right) e^{-i \frac{E_k}{\hbar} t} |k\rangle$$

21. Fermijavo zlato pravilo

$$V(t) = \begin{cases} 0, & t < 0 \\ V, & t \geq 0 \end{cases}, \quad |\psi(0)\rangle = |m\rangle, \quad P(|\psi(t)\rangle = |k\rangle) = ?$$

$$V_{km} = \langle k | V | m \rangle = \text{konst.}, \quad c_k^{(1)}(t) = \frac{1}{i\hbar} \int_0^t V_{km} e^{-i \frac{E_m - E_k}{\hbar} t'} dt' = \frac{V_{km}}{i\hbar} \frac{e^{-i\omega_{mk}t} - 1}{-i\omega_{mk}}$$

$$\hbar \omega_{mk} = E_m - E_k. \quad P_{km} = \frac{|V_{km}|^2}{\hbar^2} \frac{|e^{-i\omega_{mk}t} - 1|^2}{\omega_{mk}^2} = \frac{|V_{km}|^2}{\hbar^2} \frac{\sin^2\left(\frac{1}{2}\omega_{mk}t\right)}{\left(\frac{1}{2}\omega_{mk}\right)^2}$$

$$* \delta_t = \frac{1}{\pi} \frac{\sin^2(xt)}{x^2 t} \xrightarrow{t \rightarrow \infty} \delta(x)$$

$$\rightarrow P_{km} = \frac{2\pi}{\hbar} |V_{km}|^2 \delta_t(E_k - E_m) t$$

$$\pi \delta_t\left(\frac{E_k - E_m}{2\hbar}\right) t$$

gostota stanj $g(E_k) = \frac{dN_{stanj}}{dE_k}$, $P(t) = \int P_{km}(t) g(E_k) dE_k$

→ po dolgem času $P(t) = \frac{2\pi}{\hbar} |V_{km}|^2 g(E_m) t$

⇒ $w_{km} = \frac{dP}{dt} = \frac{2\pi}{\hbar} |V_{km}|^2 g(E_m)$ hitrost naraščanja

22. Harmonski oscilator: $a, a^\dagger$, ~~lastni~~ lastni vektorji $|n\rangle, \dots$

$$H = \frac{p^2}{2m} + \frac{1}{2} k x^2 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2, \quad \omega^2 = \frac{k}{m}$$

$$H = \frac{1}{2} \hbar \omega \left(\frac{x^2}{\xi^2} - \xi^2 \frac{d^2}{dx^2} \right), \quad \xi^2 = \frac{\hbar}{m \omega} \quad \text{karakteristična dolžina}$$

kar. energija

$$H = \frac{1}{4} \hbar \omega \left(\left(\frac{x}{\xi} + \xi \frac{d}{dx} \right) \left(\frac{x}{\xi} - \xi \frac{d}{dx} \right) + \left(\frac{x}{\xi} - \xi \frac{d}{dx} \right) \left(\frac{x}{\xi} + \xi \frac{d}{dx} \right) \right)$$

→ uvedemo $a = \frac{1}{\sqrt{2}} \left(\frac{x}{\xi} + \xi \frac{d}{dx} \right)$, $a^\dagger = \frac{1}{\sqrt{2}} \left(\frac{x}{\xi} - \xi \frac{d}{dx} \right)$

→ $x = \frac{\xi}{\sqrt{2}} (a + a^\dagger)$, $\frac{d}{dx} = \frac{1}{\sqrt{2} \xi} (a - a^\dagger)$

z uporabo $[x, p] = i\hbar$ dobimo $[a, a^\dagger] = 1$

⇒ $H = \frac{1}{2} \hbar \omega (a a^\dagger + a^\dagger a) = \hbar \omega \left(a^\dagger a + \frac{1}{2} \right) = \hbar \omega \left(\hat{n} + \frac{1}{2} \right)$

Lastne vrednosti in funkcije

$$H|\varphi_\lambda\rangle = \lambda|\varphi_\lambda\rangle \quad \langle\varphi_\lambda|H|\varphi_\lambda\rangle = \underbrace{\langle a\varphi_\lambda|a\varphi_\lambda\rangle}_{\geq 0} = \lambda\langle\varphi_\lambda|\varphi_\lambda\rangle \geq 0 \Rightarrow \lambda \geq 0$$

$a|\varphi_0\rangle = 0$ Ali ∃? $\xi \frac{d}{dx} \varphi_0(x) = -\frac{x}{\xi} \varphi_0(x) \Rightarrow \varphi_0(x) = \frac{1}{\sqrt{\sqrt{\pi} \xi}} e^{-\frac{1}{2} \left(\frac{x}{\xi} \right)^2} = \langle x|\varphi_0\rangle$

$$[n, a^\dagger] = [a^\dagger a, a^\dagger] = a^\dagger \underbrace{[a, a^\dagger]}_1 + \cancel{[a^\dagger, a^\dagger]a} = a^\dagger$$

$$n a^\dagger |\varphi_\lambda\rangle = (a^\dagger n + a^\dagger) |\varphi_\lambda\rangle = (\lambda+1) a^\dagger |\varphi_\lambda\rangle \Rightarrow a^\dagger |\varphi_\lambda\rangle = |\varphi_{\lambda+1}\rangle$$

$$\text{velja } \langle \varphi_\lambda | \varphi_\lambda \rangle = 1 \Rightarrow \langle a^\dagger \varphi_\lambda | a^\dagger \varphi_\lambda \rangle = \langle \varphi_\lambda | a a^\dagger \varphi_\lambda \rangle = \dots = \langle \varphi_\lambda | (\lambda+1) \varphi_\lambda \rangle$$

$$\Rightarrow |\varphi_{n+1}\rangle = \frac{a^\dagger}{\sqrt{n+1}} |\varphi_n\rangle, \quad |\varphi_n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}} |\varphi_0\rangle$$

$$[n, a] = \dots = -a, \quad n a |\varphi_n\rangle = (a n - a) |\varphi_n\rangle = (n-1) a |\varphi_n\rangle$$

$$\Rightarrow |\varphi_n\rangle = \frac{a}{\sqrt{n+1}} |\varphi_{n+1}\rangle, \quad |\varphi_0\rangle = \frac{a^n}{\sqrt{n!}} |\varphi_n\rangle$$

23. Ehrenfestov teorem

$$\frac{d\langle p \rangle}{dt} = \cancel{\frac{d\langle p \rangle}{dt}} + \frac{1}{i\hbar} \langle [p, H] \rangle$$

$$[p, H]f = \cancel{\frac{1}{2m} [p, p^2]} f + [p, V(x)]f = i\hbar V \frac{df}{dx} - i\hbar \frac{\partial(Vf)}{\partial x} = -i\hbar \left(\frac{\partial V}{\partial x}\right)f$$

$$\Rightarrow \frac{d\langle p \rangle}{dt} = \left\langle -\frac{\partial V}{\partial x} \right\rangle, \text{ enako za } y \text{ in } z \Rightarrow \underline{\frac{d\langle \vec{p} \rangle}{dt} = \langle -\nabla V \rangle = \langle \vec{F} \rangle}$$

$$\text{Velja tudi: } \frac{d\langle \vec{L} \rangle}{dt} = \langle \vec{M} \rangle. \text{ Za dokaz se uporabi } [x_\alpha, x_\beta] = [p_\alpha, p_\beta] = 0, \\ [x_\alpha, p_\beta] = i\hbar \delta_{\alpha\beta}$$

24. Vrtilna količina: lastnosti, baza

$$\vec{L} = \vec{L}^\dagger = \vec{r} \times \vec{p} = -\vec{p} \times \vec{r}$$

$$\begin{aligned} [L_x, L_y] &= [y p_z - p_y z, z p_x - x p_z] = [y p_z, z p_x] - \cancel{[p_y z, z p_x]} - \cancel{[y p_z, x p_z]} + \\ &+ [p_y z, x p_z] = y p_x [p_z, z] + x p_y [z, p_z] = i\hbar (x p_y - y p_x) = i\hbar L_z \end{aligned}$$

$$U(\varphi) A = A U(\varphi) \Rightarrow [L, A] = 0$$

$$[L_x, L^2] = 0$$

↳ rotacija za φ

$$L_{\pm} = L_x \pm iL_y, \quad [L^2, L_{\pm}] = 0$$

$$[L_z, L_{\pm}] = \underbrace{[L_z, L_x]}_{i\hbar L_y} \pm i \underbrace{[L_z, L_y]}_{-i\hbar L_x} = \pm \hbar L_{\pm}$$

$$L_{\pm} L_{\mp} = \dots = L^2 - L_z^2 \pm \hbar L_z \Rightarrow [L_+, L_-] = 2\hbar L_z$$

Lastne vrednosti: $L_z |m\rangle = m\hbar |m\rangle$

$$L_z L_{\pm} |m\rangle = (L_{\pm} L_z \pm \hbar L_{\pm}) |m\rangle = (m \pm 1)\hbar L_{\pm} |m\rangle \propto |m \pm 1\rangle$$

$$L^2 |m\rangle = \lambda |m\rangle \rightarrow \langle m | L^2 | m \rangle = \lambda \underbrace{\langle m | m \rangle}_1 = \sum_{\alpha} \langle L_{\alpha} m | L_{\alpha} m \rangle \geq 0$$

$$\Rightarrow \lambda \geq 0 \quad \text{vedemo } l: \lambda = \hbar^2 l(l+1)$$

$$0 \leq \langle L_{\pm} \psi_m | L_{\pm} \psi_m \rangle = \langle m | L_{\mp} L_{\pm} | m \rangle = \langle m | (L^2 - L_z^2 \mp \hbar L_z) | m \rangle =$$

$$= \langle m | (l(l+1) - m(m \pm 1)) \hbar^2 | m \rangle = C_{l, m \pm 1}^2 \langle m | m \rangle$$

$$\Rightarrow C_{l, m \pm 1} = \hbar \sqrt{l(l+1) - m(m \pm 1)} \in \mathbb{R} \Rightarrow l(l+1) \geq m(m \pm 1) \Rightarrow |m| \leq l$$

Spekter L^2 je $(2l+1)$ -krat degeneriran, $l = \begin{cases} 0, 1, 2, \dots \\ \frac{1}{2}, \frac{3}{2}, \dots \end{cases}$

$$m = -l, -(l+1), \dots, l$$

Lastne funkcije so sferični harmoniki $Y_l^m(\vartheta, \varphi)$

25. Teorija motenj, degeneriran 1. red

$$N=2 \text{ (lastna stanja } H_0 \text{ so } 2 \times \text{deg.)} \quad H = H_0 + \lambda V$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \dots, \quad |n\rangle = c_1 |n_1^0\rangle + c_2 |n_2^0\rangle + \dots \lambda |n^1\rangle + \dots$$

$$\lambda^0: H_0 |n_1^0\rangle = E_n^{(0)} |n_1^0\rangle, \quad H_0 |n_2^0\rangle = E_n^{(0)} |n_2^0\rangle$$

$$\lambda^1: H_0 |n^1\rangle + c_1 V |n_1^0\rangle + c_2 V |n_2^0\rangle = E_n^{(0)} |n^1\rangle + E_n^{(1)} (c_1 |n_1^0\rangle + c_2 |n_2^0\rangle)$$

$$(i) / \langle n_1^0 | : \quad \cancel{\langle n_1^0 | H_0 | n^1 \rangle} + \underbrace{c_1 \langle n_1^0 | V | n_1^0 \rangle}_{V_{11}} + \underbrace{c_2 \langle n_1^0 | V | n_2^0 \rangle}_{V_{12}} = \cancel{E_n^{(0)} \langle n_1^0 | n^1 \rangle} + E_n^{(1)} (c_1 \underbrace{\langle n_1^0 | n_1^0 \rangle}_1 + \cancel{c_2 \langle n_1^0 | n_2^0 \rangle})$$

$$c_1 V_{11} + c_2 V_{12} = E_n^{(1)} c_1$$

$$(ii) / \langle n_2^0 | \Rightarrow \dots \quad c_1 V_{21} + c_2 V_{22} = E_n^{(1)} c_2$$

$$\rightarrow \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = E_n^{(1)} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \Rightarrow \det(V_{ij} - E_n^{(1)} \delta_{ij}) = 0$$

$$\rightarrow E_n^{(1)}, c_{iL} \quad \checkmark$$

26. Seštevanje VK $1/2 \times 1/2$: singlet, triplet

$$S_n^2 |s_n m_n\rangle = s_n(s_n+1)\hbar^2 |s_n m_n\rangle, \quad S_{nz} |s_n m_n\rangle = m_n \hbar |s_n m_n\rangle, \quad n=1,2$$

$$[S_{n\alpha}, S_{n\beta}] = i\hbar \epsilon_{\alpha\beta\gamma} S_{n\gamma} \delta_{nn}, \quad \vec{S}_n = \frac{\hbar}{2} \vec{\sigma}_n, \quad \langle \psi | \vec{S} | \psi \rangle = \sum_n \langle \chi_n | \vec{S}_n | \chi_n \rangle$$

$$\vec{S} = \vec{S}_1 \otimes I_2 + I_1 \otimes \vec{S}_2, \quad |m_1 m_2\rangle = |m_1\rangle \otimes |m_2\rangle \quad (|s_n m_n\rangle = \frac{1}{2} |m_n\rangle = |m_n\rangle) \quad \downarrow \text{spinor}$$

$$S_z |m_1 m_2\rangle = (S_{1z} \otimes I_2 + I_1 \otimes S_{2z}) |m_1\rangle \otimes |m_2\rangle = \hbar m_1 |m_1\rangle \otimes |m_2\rangle + |m_1\rangle \hbar m_2 |m_2\rangle = (m_1 + m_2) \hbar |m_1 m_2\rangle$$

$$[S_\alpha, S_\beta] = [S_{1\alpha} \otimes I_2 + I_1 \otimes S_{2\alpha}, S_{1\beta} \otimes I_2 + I_1 \otimes S_{2\beta}] = \underbrace{[S_{1\alpha}, S_{1\beta}] \otimes I_2}_{i\hbar \epsilon_{\alpha\beta\gamma} S_{1\gamma}} + I_1 \otimes [S_{2\alpha}, S_{2\beta}]$$

$$= i\hbar \epsilon_{\alpha\beta\gamma} S_\gamma$$

$$S_\pm = S_x \pm i S_y = S_{1\pm} \otimes I_2 + I_1 \otimes S_{2\pm}$$

Uvedemo bazo $|sm\rangle = \sum_{m_1, m_2} c_{m_1, m_2} |m_1 m_2\rangle$, $m = m_1 + m_2$, $m_i = \pm \frac{1}{2}$

$$S^2 |sm\rangle = s(s+1)\hbar^2 |sm\rangle, \quad S_z |sm\rangle = \hbar m |sm\rangle$$

• $s=1, m=1$: $S^2 |sm\rangle = S^2 |11\rangle = 1(1+1)\hbar^2 |11\rangle = 2\hbar^2 |\uparrow\uparrow\rangle$

$$S_- |11\rangle = \hbar \sqrt{1(1+1) - 1(1-1)} |10\rangle = \sqrt{2}\hbar |10\rangle$$

$$(S_{1-} \otimes I_2 + I_1 \otimes S_{2-}) |\uparrow\uparrow\rangle = \hbar |\downarrow\uparrow\rangle \otimes |\uparrow\rangle + \hbar |\uparrow\rangle \otimes |\downarrow\rangle$$

$$\Rightarrow |10\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

ponovimo se enkrat $\rightarrow |1-1\rangle = |\downarrow\downarrow\rangle$

$|11\rangle, |10\rangle, |1-1\rangle$ so tripletna stanja

Singlet: $|00\rangle = c_{\frac{1}{2}-\frac{1}{2}} |\uparrow\downarrow\rangle + c_{-\frac{1}{2}\frac{1}{2}} |\downarrow\uparrow\rangle \perp |10\rangle \Rightarrow \langle 10|00\rangle = 0$
 $\hookrightarrow$ ker različni s

$$\Rightarrow c_{\frac{1}{2}-\frac{1}{2}} = -c_{-\frac{1}{2}\frac{1}{2}} \Rightarrow |00\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

27. Variacijska metoda določanja energije

1) $H|n\rangle = E_n |n\rangle$, E_n in $|n\rangle$ neznan, $\langle m|n\rangle = \delta_{mn}$

$$|\psi\rangle = \sum c_n |n\rangle \rightarrow \langle \psi | H | \psi \rangle = \langle \psi | \sum_n |m\rangle \langle m| H \sum_n |n\rangle \langle n| \psi \rangle = \sum_n E_n |c_n|^2$$

$$\sum_n E_n |c_n|^2 \geq E_0 \sum_n |c_n|^2 \Rightarrow \langle \psi | H | \psi \rangle \geq E_0 \sum_n |c_n|^2 = E_0 \langle \psi | \psi \rangle$$

$$\Rightarrow E_0 \leq \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$$

$$\Rightarrow \text{za ta } \psi \text{ izračunamo } \langle H \rangle = \sum (\alpha_1, \dots, \alpha_N)$$

$$\rightarrow \text{najdemo minimum energije } \sum_{\min} \Rightarrow E_0 \leq \sum_{\min}$$

$\rightarrow$ Ustvarimo ψ , ki je čim boljši približek osn. st. in je odv. od N parametrov: $\psi = \psi(\alpha_1, \dots, \alpha_N, x)$

2) Perturbacije $H = H_0 + \lambda V$, $H_0 |\varphi_n^0\rangle = E_n^{(0)} |\varphi_n^0\rangle$

$$\rightarrow E_0 \leq \langle \psi_0 | H | \psi_0 \rangle = E_0^{(0)} + \lambda \langle \varphi_0^0 | V | \varphi_0^0 \rangle$$

28. Delov v ~~mag~~ EMP, magn. moment, Zeemanov pojav

$$H = \frac{(p - eA)^2}{2m} + e\phi(\vec{r}), \quad \vec{B} = \nabla \times \vec{A}, \quad E = -\nabla\phi - \frac{\partial A}{\partial t}$$

$$\begin{aligned} SE: \quad i\hbar \frac{\partial \psi}{\partial t} &= \frac{1}{2m} (-i\hbar \nabla - e\vec{A})^2 \psi + e\phi \psi = \\ &= -\frac{\hbar^2 \nabla^2}{2m} \psi + i \frac{\hbar e}{m} \vec{A} \cdot \nabla \psi + \left(i \frac{\hbar e}{2m} \nabla \cdot \vec{A} + \frac{e^2 A^2}{2m} + e\phi \right) \psi \end{aligned}$$

Običajno $\vec{A}$ umerimo tako, da je $\nabla \cdot \vec{A} = 0$.

Zeemanov sklopitev: $\vec{A} = \frac{1}{2} \vec{B} \times \vec{r}$, $\nabla \cdot \vec{A} = 0$, $\vec{B} = (0, 0, B)$

$$i \frac{\hbar e}{m} \vec{A} \cdot \nabla \psi = -i \frac{\hbar e}{2m} (\vec{r} \times \vec{B}) \cdot \nabla \psi = i \frac{\hbar e}{2m} (\vec{r} \times \nabla) \cdot \vec{B} \psi = -\frac{e}{2m} \vec{B} \cdot \vec{L} \psi = -\vec{B} \cdot \vec{\mu} \psi$$

μ ... magnetni moment, $\mu_B = \frac{|e|\hbar}{2m_e}$ Bohrov magneton

$\rightarrow H_{\text{Zeeman}} = -\vec{\mu} \cdot \vec{B}$ normalna Zeemanova skl.
 $\vec{B} = (0, 0, B)$

$$\frac{e^2}{2m} A^2 = \frac{e^2}{8m} (B^2 r^2 - (B \cdot r)^2) \stackrel{!}{=} \frac{e^2 B^2}{8m} (x^2 + y^2)$$

$$\Rightarrow H = \frac{p^2}{2m} - \vec{\mu} \cdot \vec{B} + V, \quad V = e\phi + \frac{e^2 B^2}{8m} (x^2 + y^2)$$

$$H|nlm\rangle = \left(-\frac{R_y}{n^2} + m\mu_B B \right) |nlm\rangle$$

29. Schrödingerjeva enačba in spin, Heisenbergova sklopitev

$$\text{Heisenberg: } H = J_0 \vec{S}_1 \cdot \vec{S}_2, \quad S_{1,2} = \frac{1}{2}$$

$\hookrightarrow$ izmenjalna sklopitvena konstanta

$J_0 < 0 \rightarrow$ feromagn. skl., $\vec{V}_{K,2}$ sta vzporedni

$J_0 > 0 \rightarrow$ antiferomagn. skl., $\vec{V}_{K,2}$ sta nasprotno obrnjeni

$$\vec{S} \cdot \vec{S} = (\vec{S}_1 + \vec{S}_2) \cdot (\vec{S}_1 + \vec{S}_2) = S_1^2 + S_2^2 + 2\vec{S}_1 \cdot \vec{S}_2$$

$$\Rightarrow H = \frac{J_0}{2} \left(S^2 - 2 \cdot \hbar^2 \frac{1}{2} \left(\frac{1}{2} + 1 \right) \right) = \frac{J_0}{2} \left(S^2 - \frac{3}{2} \hbar^2 \right)$$

$$E_S = J_0 \hbar^2 \begin{cases} \frac{1}{4} & S=1 \text{ (triplet)} \\ -\frac{3}{4} & S=0 \text{ (singlet)} \end{cases}$$

Hamiltonian za dvonivojski sistem: $H = E_0 I + \sum_{\alpha} E_{\alpha} \sigma_{\alpha}$, $E_{\alpha} \in \mathbb{R}$

30. Aharonov-Bohmov pojav

v nekem delu prostora: $\vec{B} = \nabla \times \vec{A} = 0 \Rightarrow \vec{A} = \nabla \Lambda$
 $\vec{A}$ ni polje

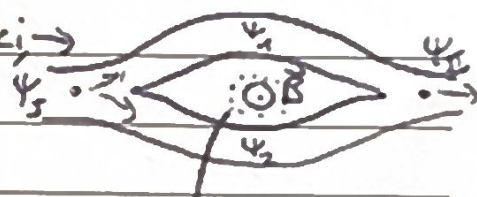
$$\Lambda(\vec{r}) = \Lambda(r_0) + \int_{r_0}^{\vec{r}} \vec{A}(\vec{r}') \cdot d\vec{r}'$$

$$\rightarrow \text{SE: } i\hbar \frac{\partial}{\partial t} \Psi = \frac{(\vec{p} - e\vec{A})^2}{2m} \Psi + V\Psi \quad \left| \quad i\hbar \frac{\partial}{\partial t} \Psi' + \frac{\vec{p}^2}{2m} \Psi' + V'\Psi', \text{ ker ni polje} \right.$$

$$\Lambda \neq \Lambda(t) \Rightarrow V = V', \quad A' = A + \nabla(-\Lambda) = 0$$

$$\Rightarrow \Psi' = \exp(-i \frac{e}{\hbar} \Lambda) \Psi \rightarrow \Psi_A(\vec{r}) = \exp(i \frac{e}{\hbar} \int_{r_0}^{\vec{r}} \vec{A}(\vec{r}') \cdot d\vec{r}') \Psi_0(r)$$

$\rightarrow$ Miselni eksperiment: valovni paket se giba po žici, ki se razcepi ($\Psi_I = \Psi_1 + \Psi_2$) na 2 dele.



e^- se gibajo po območju, kjer $B=0$ in $A \neq 0$.

$$\rightarrow \Psi_{II,B} = e^{i\delta_1} \Psi_1 + e^{i\delta_2} \Psi_2, \quad \delta_i = \frac{e}{\hbar} \int_{I(poi)}^{\vec{r}} \vec{A}(\vec{r}') \cdot d\vec{r}'$$

$\vec{B} \neq 0$ znotraj kroga

Če je naprava simetrična: $\Psi_{II,B} = e^{i\delta_2} (1 + e^{i(\delta_1 - \delta_2)}) \Psi_1$,

$$\delta_1 - \delta_2 = \frac{e}{\hbar} \oint_{\text{Stokes } S} \vec{A} d\vec{r} = \frac{e}{\hbar} \int_S \nabla \times \vec{A} \cdot d\vec{S} = \frac{e}{\hbar} \int_S \vec{B} \cdot d\vec{S} = \frac{e}{\hbar} \Phi_B$$

$\rightarrow$ tok ob odsotnosti $\vec{B}$

A-B pojav: razmerje med I_B in I_0 :

$$\frac{I_B}{I_0} = \frac{|\Psi_{II,B}|^2}{|\Psi_{II,0}|^2} = \frac{1}{4} |1 + e^{i \frac{e}{\hbar} \Phi_B}|^2 = \cos^2 \frac{e}{2\hbar} \Phi_B$$

$$\text{Maksimumi: } \frac{e}{2\hbar} \Phi_B = n\pi \Rightarrow \Phi_B = n \frac{h}{e}$$

31. Berryjeva faza

~~$$H \in H^2(U), \frac{d\psi}{dt} = \frac{1}{i\hbar} (E_n - E_n) \psi$$~~

$$\vec{Q} = (q_1, q_2, \dots, q_N), H = H(Q(t))$$

$$H(Q) |\psi_n(Q)\rangle = E_n(Q) |\psi_n(Q)\rangle \text{ nedeg.}$$

razen

$$|\psi_n\rangle = e^{i\phi_n(t)} |\psi_n^0\rangle, \quad i\hbar \frac{\partial}{\partial t} |\psi_n\rangle = H |\psi_n\rangle$$

$$\rightarrow i\hbar \left(i \frac{d\phi_n}{dt} e^{i\phi_n} |\psi_n^0\rangle + e^{i\phi_n} \frac{\partial}{\partial t} |\psi_n^0\rangle \right) = H e^{i\phi_n} |\psi_n^0\rangle = E_n e^{i\phi_n} |\psi_n^0\rangle \quad / \cdot \langle \psi_n^0 |$$

$$i\hbar \left(i \frac{d\phi_n}{dt} \langle \psi_n^0 | \psi_n^0 \rangle + \langle \psi_n^0 | \frac{\partial}{\partial t} |\psi_n^0\rangle \right) = E_n \langle \psi_n^0 | \psi_n^0 \rangle$$

Zapišemo $\phi_n = \Theta_n + \gamma_n$, da velja $i\hbar \left(i \frac{d\gamma_n}{dt} + \langle \psi_n^0 | \frac{\partial}{\partial t} |\psi_n^0\rangle \right) = \underbrace{E_n + \hbar \frac{d\Theta_n}{dt}}_0$

$$\rightarrow \Theta_n = -\frac{1}{\hbar} \int_0^t E_n(t') dt', \quad \frac{d\gamma_n}{dt} = i \langle \psi_n^0 | \frac{\partial}{\partial t} |\psi_n^0\rangle$$

$$\frac{\partial \psi_n^0}{\partial t} = \sum_{k=1}^N \frac{\partial \psi_n^0}{\partial q_k} \dot{q}_k = (\nabla_Q \psi_n^0) \cdot \dot{Q} \rightarrow \gamma_n(t) = \int_0^t i \langle \psi_n^0 | \nabla_Q \psi_n^0 \rangle \cdot \underbrace{\dot{Q} dt}_{dQ}$$

$$\rightarrow \phi_n(t) = \underbrace{-\frac{1}{\hbar} \int_0^t E_n(t) dt}_{\text{dinamična faza}} + \underbrace{\int_{Q(0)}^{Q(t)} i \langle \psi_n | \nabla_Q \psi_n \rangle \cdot dQ}_{\text{geometrijska faza}}$$

$$A_n(Q) = i \langle \psi_n | \nabla_Q \psi_n \rangle$$

Berryjeva ~~faza~~ povezava

$$\nabla_Q \underbrace{\langle \psi_n | \psi_n \rangle}_1 = 0 = \langle \nabla_Q \psi_n | \psi_n \rangle + \langle \psi_n | \nabla_Q \psi_n \rangle = 2 \operatorname{Re} \langle \psi_n | \nabla_Q \psi_n \rangle = 2 \operatorname{Re} A_n$$

$$\Rightarrow \text{če } \psi_n \in \mathbb{R} \Rightarrow A_n = 0$$

Če je trajektorija ob t zaključena:

$H(Q(0)) = H(Q(t))$, se stanje pomnoži z dinamično in Berryjevo fazo

$$\gamma_{\text{Berry}} = \oint A_n(Q) \cdot dQ$$

Berryjeva ukrivljenost:

$$\Omega_n(Q) = \nabla_Q \times A_n(Q)$$

$\rightarrow$ Stokes

$$\gamma_{\text{Berry}} = \int_S \Omega_n(Q) \cdot dS$$

Če S zaključena, velja $\gamma_n = 2\pi n$ (Chernov število)